

Assignment 1

Group Number 11

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1 Part A: Research Proposal

1.1 Context

As embodied AI develops, robots are increasingly expected not only to execute predefined commands but also to acquire new skills for physical tasks. This trend was clearly reflected at the 2026 World Artificial Intelligence Conference (WAIC 2026), where more than 300 embodied-intelligence robots demonstrated capabilities such as object manipulation, assembly, and collaboration [1]. This raises a practical question: when a robot needs to learn a new task, what is the most effective way to communicate the task to it?

One common approach is to provide instructions in language. However, describing a physical task in text may not fully capture the sequence, timing, and execution of actions. Learning from Demonstration (LfD) offers an alternative in which robots learn task knowledge from examples of human behaviour [2,3]. A human demonstration can provide information about how a task is actually performed, rather than only describing the desired outcome.

However, the availability of additional information does not necessarily mean that a robot will perform a task more efficiently. It is therefore interesting to test whether adding a human demonstration to a text instruction actually improves task performance compared with text instructions alone. A randomized experiment can provide a causal test of this effect by keeping the task, robot, and text instruction constant while varying only whether a demonstration is provided.

1.2 Research Question

Does adding a human demonstration to a text instruction affect the time an embodied AI robot needs to complete a manipulation task?

1.3 Hypotheses

Alternative hypothesis (H_1): Adding a human demonstration to a text instruction affects mean task completion time.

Null hypothesis (H_0): Adding a human demonstration to a text instruction has no effect on mean task completion time.

We expect the demonstration condition to result in shorter completion times.

1.4 Study Design

We propose a simulated randomized controlled experiment with 100 independent robot trials. The trials will be randomly assigned to two groups of equal size. In the text-only condition ($n = 50$), the robot receives a standardized textual instruction describing how to perform the manipulation task. In the text + demonstration condition ($n = 50$), the robot receives the same textual instruction together with a recorded human demonstration of the task. The robot model, hardware, task, environment, and text instruction will be identical across conditions. The presence of the human demonstration is therefore the only manipulated factor.

1.5 Variables and Sample

The independent variable is **instruction condition**, a nominal categorical variable with two possible values: text-only and text + demonstration. The dependent variable is **task completion time**, measured in seconds. It is a continuous ratio-level variable with a theoretical minimum of zero seconds and positive observed values. The total sample size is $N = 100$, with 50 trials per condition.

1.6 Analysis Pipeline

First, we will calculate the mean and standard deviation of task completion time for each condition and inspect the distributions for extreme values. We will then compare the two groups using an **independent-samples two-sided t -test**, with instruction condition as the grouping variable and task completion time as the outcome. The significance level will be set to $\alpha = 0.05$. We will report the mean difference between conditions, its confidence interval, and the corresponding p -value. A statistically significant difference will be interpreted as evidence that the instruction condition affects task completion time. Although we expect the demonstration condition to result in shorter completion times, the two-sided test allows for an effect in either direction.

2 Part B: Synthetic Data

2.1 Simulation of The Data

We simulated data for 100 robot trials, with 50 trials in each instruction condition. The two conditions were Text-only and Text + demonstration. The outcome variable was task completion time measured in seconds.

For the simulation, completion times were drawn from normal distributions. We assumed the Text-only condition had a baseline mean completion time of 30 seconds and a standard deviation of 5 seconds. These values were chosen as plausible values for a short robotic manipulation task and are simulation assumptions rather than estimates taken from empirical data. For the dataset with a true effect, we chose the true effect of -3 seconds, meaning that the expected completion time in the Text + demonstration condition was 27 seconds. This is because we believed a true effect of -3 seconds is a plausible and moderate effect that large enough to be visible in the simulated data while still allowing substantial overlap between the groups. Thus, the simulated population means were 30 seconds for the Text-only condition and 27 seconds for the Text + demonstration condition.

For the dataset under the null hypothesis, the effect size was set to 0 seconds. Therefore, both conditions had an expected mean completion time of 30 seconds.

A fixed random seed was used to make the simulation reproducible. Completion times were constrained to be non-negative using truncation at zero, because negative completion times are not physically possible. No rounding was applied because completion time was treated as a continuous variable. Given the chosen mean and standard deviation, negative values were extremely unlikely.

2.2 Simulated Data with A True Effect

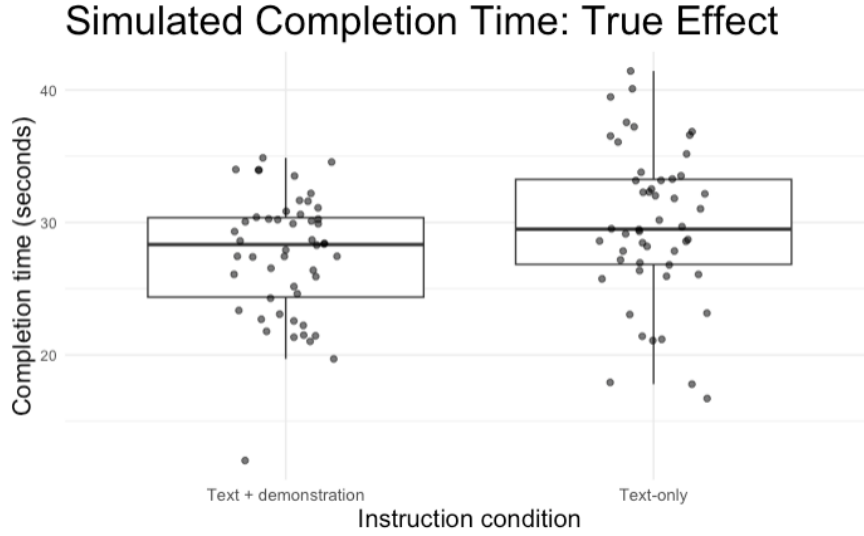


Figure 1: Simulated completion time, true-effect dataset.

Figure 1 shows the simulated completion times when a true effect of -3 seconds was assumed. The Text + demonstration condition has a lower distribution of completion times than the Text-only condition. The sample mean was 27.50 seconds for the Text + demonstration condition and 29.82 seconds for the Text-only condition, corresponding to an observed mean difference of approximately **-2.32 seconds**.

There is still substantial overlap between the two groups because completion times were simulated with a standard deviation of 5 seconds. This means that individual robots can perform faster or slower even when the demonstration has a true effect. Sampling variability also means that the observed difference in sample means does not have to equal the true effect of -3 seconds exactly.

The two-sided Welch's t-test resulted in a statistically significant difference, $t(93.65) = -2.22$, $p = .029$. Because $p < .05$, we reject the null hypothesis of no difference in mean completion time. The 95% confidence interval for the difference in means was $[-4.39, -0.24]$ seconds, which does not include zero. The negative mean difference indicates that the demonstration condition had shorter completion times in this dataset, which is consistent with the research expectation that adding a human demonstration leads to faster task completion.

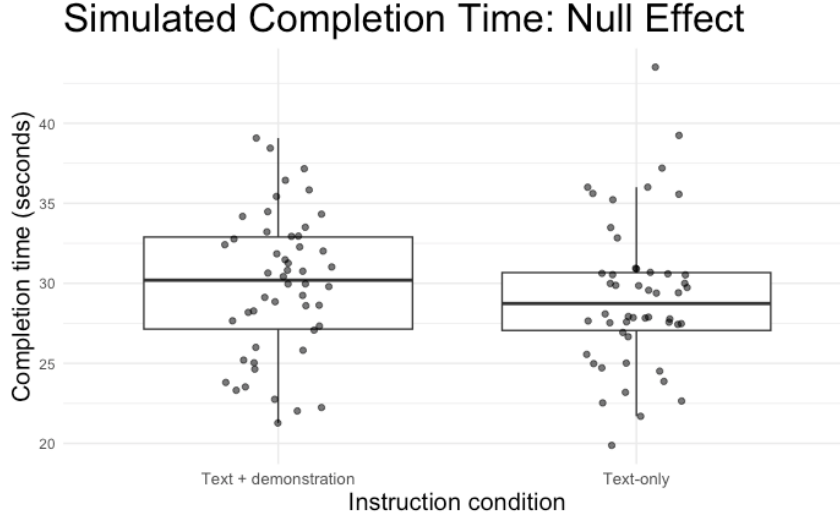


Figure 2: Simulated completion time, null dataset.

2.3 Simulated Data Under the Null Hypothesis

Figure 2 shows the simulated completion times when the true effect was set to 0 seconds. In this case, both conditions were simulated from the same normal distribution, with a population mean of 30 seconds and a standard deviation of 5 seconds.

The sample means were nevertheless not exactly equal: the Text + demonstration condition had a mean of 29.88 seconds, while the Text-only condition had a mean of 29.24 seconds. This difference is due to random sampling variation rather than a true effect.

The two-sided Welch’s t -test did not show a statistically significant difference between the conditions, $t(97.78) = 0.70$, $p = .483$. The 95% confidence interval for the difference in means was $[-1.16, 2.44]$ seconds, which includes zero. Therefore, this simulated dataset does not provide sufficient evidence to reject the null hypothesis of no difference in mean completion time.

Overall, the simulations demonstrate how the same experimental design could produce different observed data depending on whether a true effect exists. A larger separation between the two conditions would provide stronger evidence that adding a human demonstration affects task completion time.

3 Part C: Simulated analysis

3.1 Assumption Checks

Before conducting the primary analysis, the assumptions relevant to the planned comparison of completion times were assessed using the simulated data. The planned analysis was a two-sided Welch’ two-sample t -test, which compares the mean completion times between the two independent conditions without requiring equal population variances. The main distributional assumption considered was that completion times within each condition were approximately normally distributed.

The data-generating process assumed normally distributed completion times within each condition, with a standard deviation of 5 seconds.

To assess whether the realised samples were consistent with the assumed normal distribution, Q–Q plots of the model residuals were examined separately for the alternative and null datasets. For each dataset, a linear model with `completion_time` as the outcome and `condition` as the predictor was fitted, and a Q–Q plot of the model residuals was examined.

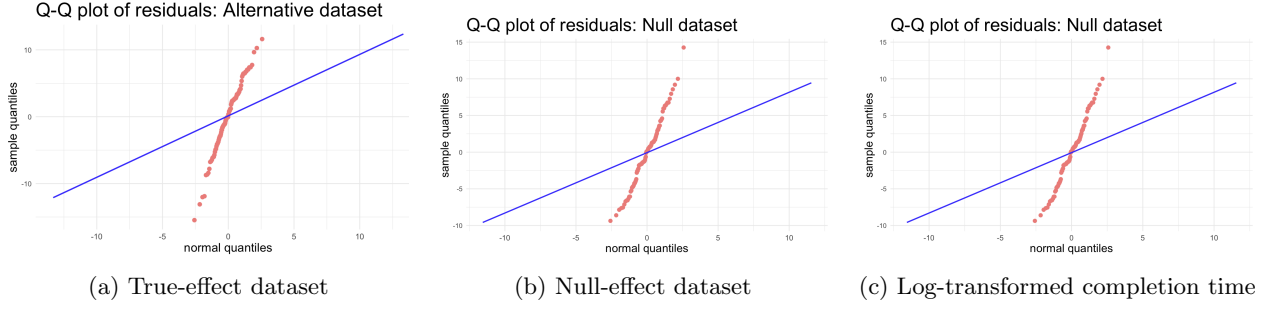


Figure 3: Q-Q plots of completion time under different data-generating conditions.

For the alternative dataset, the residual Q–Q plot showed systematic deviations from the reference line (Figure 3a). A similar pattern was observed for the null dataset (Figure 3b). These deviations indicate that the realised samples did not match the theoretical normal distribution perfectly. However, a Q–Q plot describes the particular sample that was generated and does not by itself demonstrate that the underlying data-generating distribution was non-normal. Since the simulated completion times were explicitly generated from normal distributions, the observed deviations were interpreted as variation in the realised samples rather than as evidence that the data-generating distribution should be changed.

The distributions were also examined using boxplots as shown before in Figure 1 and Figure 2. It is clear that several observations fell beyond the conventional $1.5 \times IQR$ whiskers in both conditions in both tails. These observations were not automatically treated as erroneous or removed, since observations outside the boxplot whiskers are not necessarily invalid. All observations were therefore retained.

A log transformation was also considered as a potential modification. However, the transformation did not improve the distributional diagnostics, which did not surprise us much since the distribution of the residuals in our case is less right-skewed. In particular, the histogram became more right-skewed and the Q–Q plot did not show a clear improvement (Figure 3c). The transformation was therefore not retained.

In the end, we decided in our primary analysis, no observations will be removed and no transformation will be applied to the outcome. The primary analysis is therefore retained as planned, using two-sided Welch’s t -test for both datasets. It is chosen since the research question concerns a difference in mean completion time between the two conditions. This approach also avoids selecting a different analysis solely on the basis of patterns observed in the same simulated sample used for inference.

Overall, we inspected the assumptions using the simulated datasets, where some deviations from the normal Q–Q reference line was observed. However, the data was generated from normal distributions, and deviations in a finite sample do not necessarily imply that the underlying population distribution is non-normal. Therefore, the assumption checks were treated as a diagnostic assessment and limitation rather than to search for an alternative analysis that better matched the observed data.

3.2 Simulation Design

The assumption checks suggest that the simulated data are not perfectly normal in this particular sample, as shown by the deviations of the Q–Q plot. However, this does not necessarily mean that the statistical analysis will perform poorly. Importantly, the data were generated from a known normal data-generating process using *rnorm*. Therefore, the deviations observed in one simulated dataset may simply reflect ordinary sampling variability rather than a genuine violation of the underlying model.

This motivates the next step: instead of looking at only one simulated dataset, we can try

to explore what would happen if the same study were repeated many times, with a new random sample generated each time from the same population. To answer this, we simulated **10,000** studies under both the alternative and null hypothesis. For each study, new data were generated, the same Welch t-test was conducted, and the resulting p-value and estimated effect were recorded. This allows the behaviour of the test to be examined across many possible samples, rather than drawing conclusions from a single test. And it provides a more informative assessment of how the statistical test behaves under the specified data-generating process, including its Type I error rate, Type II error rate, statistical power, and the amount of variation in estimated effects across studies.

3.3 Type I Error, Type II Error and Power

Table 1 summarises the error rates. Under the null hypothesis, 481 of the 10000 studies were significant, giving an estimated Type I error rate of 4.81%. This is close to the nominal significance level of $\alpha = 5\%$. Therefore, the simulation suggests that the planned two-sided Welch t-test approximately maintains the intended Type I error rate.

Under the alternative hypothesis, 8420 studies were significant, giving an estimated power of 84.2% and a Type II error rate of 15.8%. This agrees with the analytical power for this design (`power.t.test`: 84.2%). In other words, even though the demonstration truly reduces completion time by 3 s, roughly one in six studies with this design would fail to detect it; and the test detected the true effect in approximately 84 out of every 100 simulated studies.

Measure	Estimate
Type I error rate	0.048
Type II error rate	0.158
Statistical power	0.842

Table 1: Estimated error rates and power from 10000 simulated studies per hypothesis ($n = 50$ per group, $\alpha = .05$).

The Type II error rate and statistical power sum to 1, as expected: $\beta + Power = 0.158 + 0.842 = 1.00$.

3.4 Distributions of p-values

The simulated p-value distributions provide a visual illustration of the difference between the null and alternative hypotheses.

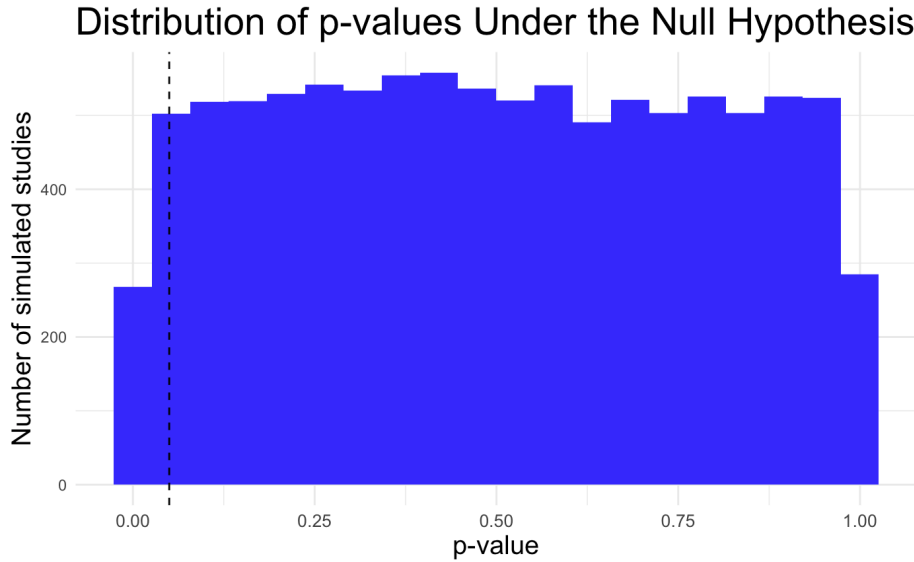


Figure 4: Distribution of p-values under the null hypothesis.

Under the null hypothesis as shown in Figure 4, the p-values should be approximately uniformly distributed between 0 and 1. This is what we expect when there is no true difference between the two conditions: all p-values are possible, although about 5% should fall below .05 significance threshold. The observed Type I error rate of 0.0481 is consistent with this expectation.

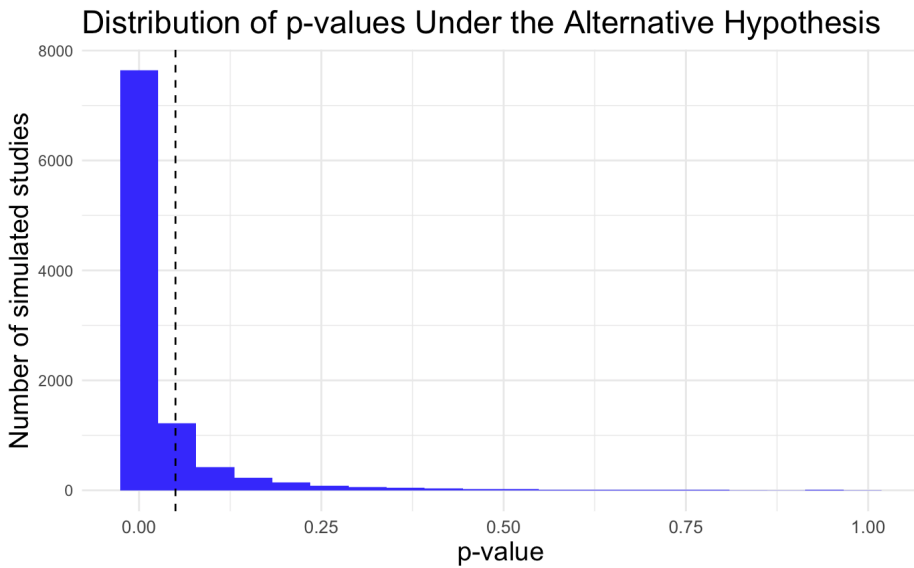


Figure 5: Distribution of p-values under the alternative hypothesis.

Under the alternative hypothesis as shown in Figure 5, the distribution of p-values is concentrated more strongly toward zero. This occurs because a true effect of -3 seconds was introduced into the data-generating process. Consequently, many simulated studies produced p-values below .05. The proportion below .05 corresponds to the estimated statistical power of 0.842.

Thus, the two p-value distributions illustrate an important distinction: under the null hypothesis, only a small proportion of studies should produce statistically significant results by chance, whereas under the alternative hypothesis, significant results should occur much more frequently because a genuine effect is present.

3.5 Variations in Estimated Effects

The simulation also demonstrates that the estimated effect is not identical from study to study, even though the true population effect remains fixed.

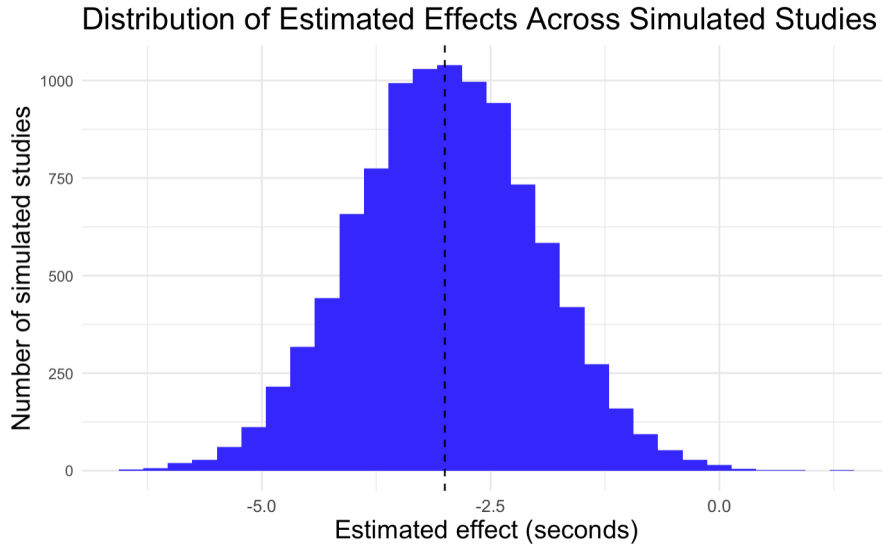


Figure 6: Distribution of estimated effects across simulated studies under the alternative hypothesis.

Under the alternative hypothesis, the true effect was -3 seconds. Across the 10,000 simulated studies, the mean estimated effect was: $M = -2.999$ seconds, which is extremely close to the true value of -3 seconds (Figure 6). This indicates that the estimator is approximately unbiased under the simulation conditions. However, the standard deviation of the estimated effects was approximately $SD = 0.999$ seconds, which shows that individual studies can produce estimates that differ substantially from the true effect. The central 95% of simulated estimated effects was approximately $[-4.96, -1.02]$ seconds. Thus, although the average estimate across many studies was almost exactly -3 seconds, individual studies could produce estimates ranging from a substantially larger negative effect to a much smaller negative effect.

This illustrates an important feature of sampling variability: a single study provides only one noisy estimate of the underlying population effect. Repeating the study many times reveals the distribution of estimates that could reasonably occur because of random sampling variation.

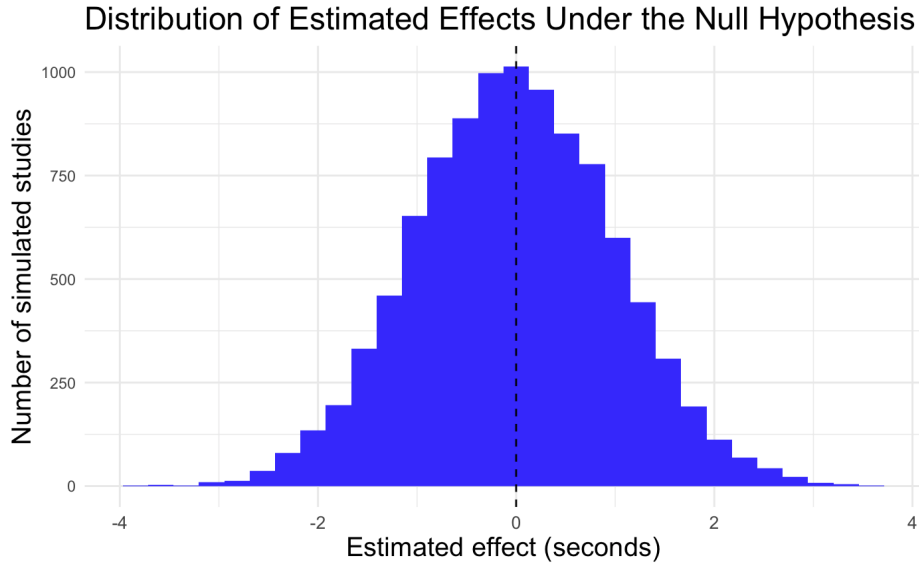


Figure 7: Distribution of estimated effects across simulated studies under the null hypothesis.

The estimated-effect distribution under the null hypothesis was centered approximately around zero (Figure 7), as expected, because the true effect in that simulation was zero.

3.6 Effect of Sample Size on Statistical Power

We further examined how sample size affects the ability to detect the true effect. Additional simulations were conducted using 2,000 studies for each of our four sample sizes per condition: 25, 100, 200, and 500.

The estimated statistical power was:

Sample size per group	Statistical power
25	0.561
100	0.988
200	1.000
500	1.000

Table 2: Estimated statistical power for different sample sizes based on 2000 simulated studies per sample size ($\alpha = .05$).

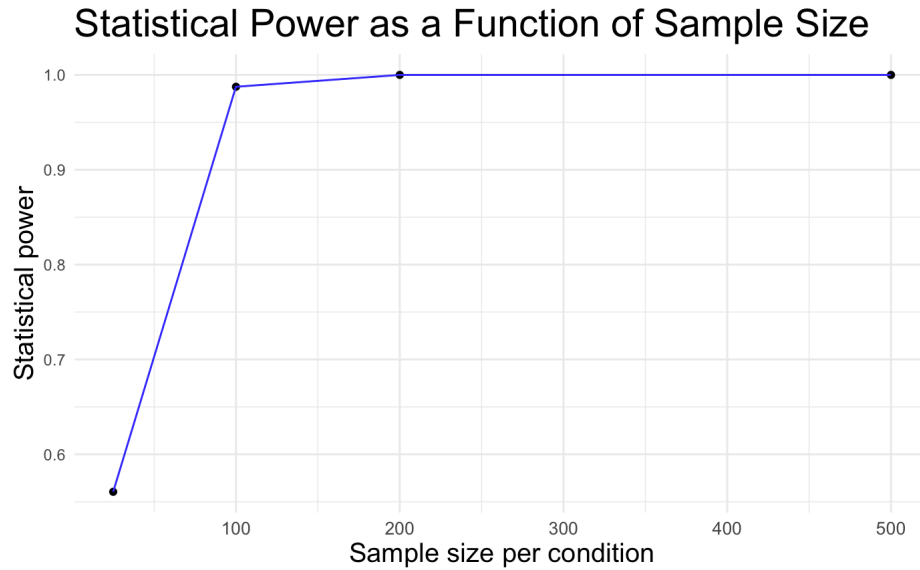


Figure 8: Distribution of estimated effects across simulated studies under the null hypothesis.

The results show a clear increase in statistical power as the sample size increases (Figure 8). With 25 participants per condition, power was approximately 56.1%, meaning that the true -3-second effect was detected in only a little more than half of the simulated studies.

Increasing the sample size to 100 participants per condition increased power substantially to 98.76%. With 200 and 500 participants per condition, the estimated power reached 100% in the 2,000 simulations conducted for each sample size.

This pattern is consistent with the general relationship between sample size and statistical power. Larger samples reduce sampling variability and therefore make it easier to distinguish a genuine effect from random variation. Importantly, the value of 1,000 does not mean that statistical power is mathematically guaranteed to be exactly 100%; rather, none of the 2,000 simulated studies failed to reach statistical significance for those sample sizes.

3.7 Conclusion

Taken together, the simulations provide a useful assessment of the expected behaviour of the planned statistical test.

First, the estimated Type I error rate of 4.81% was close to the intended 5% significance level, suggesting that the test produces false-positive results at approximately the expected rate when the null hypothesis is true.

Second, when the true effect was -3 seconds, the estimated statistical power was 84.2%, with a corresponding Type II error rate of 15.8%. Thus, under the assumptions used in the simulation, the study would detect the true effect in approximately 84% of repeated studies.

Third, the estimated effect varied considerably between simulated studies. Although the mean estimated effect was almost identical to the true effect, the standard deviation was approximately 1.00 second, and the central 95% of estimates ranged from approximately -4.96 to -1.02 seconds. This demonstrates that sampling variability can lead individual studies to overestimate or underestimate the true effect.

Finally, the additional sample-size simulation showed that statistical power increases substantially with sample size.

In summary, the simulation shows how much a single study can vary: the same experiment produces a range of estimated effects, and in about one in six cases it produces no significant result at all. A non-significant result from a single study should therefore not be read as evidence that

demonstrations have no effect.

These findings depend on the parameters of the simulation. Power increases with the true effect size and the sample size, and decreases with the within-group standard deviation, because these determine how far the distribution of estimated effects lies from zero relative to its spread. A larger sample reduces the standard error, so estimates cluster more tightly around the true value and the effect is detected more often. A stricter significance level would lower the Type I error rate but also lower power. The Type I error rate itself does not depend on sample size or effect size: as long as the assumptions of the test hold and the analysis is carried out as planned, it stays at α .

References

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